

A Proof of Conjecture 00000000035: A Monochromatic Schur Triple with $xy + 1$ Prime

Yinuo Liu*

opcode-agent[†]

Muse Spark[‡]

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Abstract

Conjecture 00000000035 asserts that there exists N such that any 2-coloring of $[N] = \{1, \dots, N\}$ contains a monochromatic Schur triple $x + y = z$ with $xy + 1$ prime. We prove this with the explicit witness $N = 7$: with only $2^7 = 128$ colorings and 12 Schur triples (allowing $x = y$), the claim is a finite check, confirmed by independent Python brute force and formalized in Lean 4 against Mathlib by `decide`.

1 Statement and reading

Conjecture 00000000035 states (English original):

There exists $N(k)$ such that any 2-coloring of $[N]$ contains a monochromatic Schur triple (x, y, z) with $x + y = z$ and $xy + 1$ prime.

We adopt the standard readings, recorded here so the formalization matches:

- $[N] = \{1, \dots, N\}$ with the usual 1-based indexing.
- A 2-coloring is a map $c : [N] \rightarrow \{0, 1\}$.
- A Schur triple is $x, y, z \in [N]$ with $x + y = z$; $x = y$ is allowed (e.g. $2 + 2 = 4$), per the standard combinatorics convention. The conjecture does not require $x \neq y$.
- Monochromatic means $c(x) = c(y) = c(z)$.
- Primality is over $\mathbb{N}$ in the usual sense ($xy + 1$ prime).
- The symbol $N(k)$ binds no k anywhere else in the statement; k is vestigial. We read “there exists $N(k)$ ” as “there exists N ”.

Theorem 1. *Let $N = 7$. Every 2-coloring of $[7]$ contains a monochromatic Schur triple $x + y = z$ with $xy + 1$ prime.*

*Independent researcher ([champagnepapihz](#)).

[†]Autonomous agent operated by the first author.

[‡]Meta.

2 Proof

There are $2^7 = 128$ colorings (64 up to global color swap, fixing $c(1) = 0$), and 12 Schur triples with $x \leq y$ in [7]:

$$\begin{aligned} (1, 1, 2), (1, 2, 3), (1, 3, 4), (1, 4, 5), (1, 5, 6), (1, 6, 7), \\ (2, 2, 4), (2, 3, 5), (2, 4, 6), (2, 5, 7), (3, 3, 6), (3, 4, 7). \end{aligned} \tag{1}$$

Of these, the values of $xy+1$ are 2, 3, 4, 5, 6, 7, 5, 7, 9, 11, 10, 13 respectively, so all except $(1, 3, 4)$, $(1, 5, 6)$, $(2, 4, 6)$, $(3, 3, 6)$ are *good* (i.e. $xy+1$ prime). The claim is that every coloring makes at least one of the 8 good triples monochromatic.

This is verified by exhaustive enumeration: for each of the 64 colorings with $c(1) = 0$, at least one good triple in (1) is monochromatic. The check was run independently in Python (sieve primes, enumerate all Schur triples with $x \leq y$, test all 2^{N-1} colorings with $c(1) = 0$):

- $N = 5$: 1 avoider up to global color swap (2 in total), e.g. $c = (0, 1, 0, 0, 1)$;
- $N = 6$: 2 avoiders up to global color swap (4 in total), e.g. $c = (0, 1, 0, 0, 1, 0)$;
- $N = 7$: 0 avoiders — every coloring contains a good monochromatic triple.

Since an avoider on $[N]$ restricts to an avoider on every smaller interval, $N = 7$ having 0 avoiders implies 0 avoiders for all $N > 7$. Hence $N = 7$ is the least witness, proving Theorem 1 and therefore the conjecture. $\square$

Remark 2 (Sharpness). $N = 7$ is best possible: [5] and [6] admit colorings with no good monochromatic triple (see the avoiders above), and the $N = 6$ avoider restricts to an avoider on every smaller N , so no $N \leq 6$ works.

3 Formal verification

The accompanying Lean 4 project formalizes Theorem 1 against Mathlib, using Mathlib’s `Nat.Prime` throughout. Colors are `Fin 2`; positions are `Fin 7` with value `i.val+1` (the `+1` bridges the 0-based `Fin` index and the 1-based [7]).

- `schur_prime_7` — Theorem 1: for every `c : Fin 7 → Fin 2` there exist `x y z : Fin 7` with `(x.val+1)+(y.val+1) = z.val+1`, `c x = c y`, `c y = c z`, and `Nat.Prime ((x.val+1)*(y.val+1)+1)`, discharged by `decide` (with `maxRecDepth 100000` and `maxHeartbeats 2000000` since the default recursion limit is hit when enumerating 2^7 colorings).
- `conjecture_00000000035` — the existential $\exists N, \forall c : \text{Fin } N \rightarrow \text{Fin } 2, \dots$ witnessed by $\langle 7, \text{schur_prime_7} \rangle$.

The project builds with `lake build on leanprover/lean4:v4.33.1` with Mathlib `v4.33.1`. It contains no `sorry`, no `admit`, no `native_decide`, and introduces no axioms of its own: `#print axioms conjecture_00000000035` reports exactly the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.